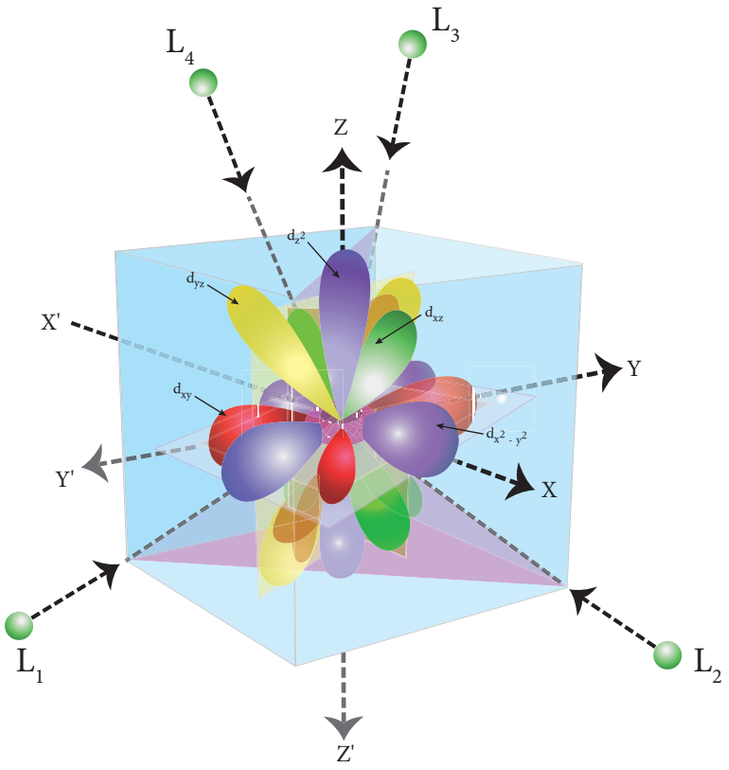
In this field, the t_{2g} orbitals (d_{xy}, d_{yz} and d_{zx}) are pointing close to the direction in which ligands are approaching than the e_g orbitals ($d_{x^2-y^2}$ and d_{z^2}).

As a result, the energy of t_2 orbitals increases by $2/5\Delta_t$ and that of e orbitals decreases by $3/5\Delta_t$ as shown below. This splitting is inverted when compared to the octahedral field.



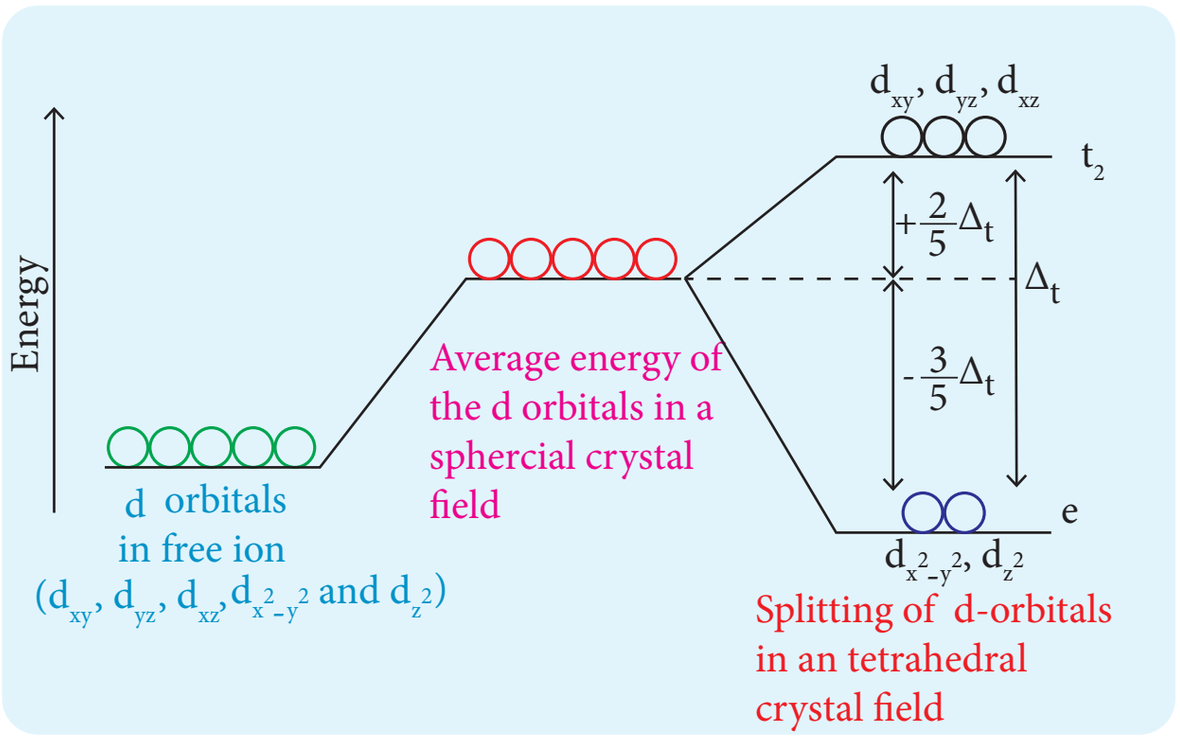


Figure: 5.13 - Crystal field splitting in tetrahedral field

Crystal filed splitting Energy and nature of ligands:

The magnitude of crystal field splitting energy not only depends on the ligand field as discussed above but also depends on the nature of the ligand, the nature of the central metal atom/ion and the charge on it. Let us understand the effect of the nature of ligand

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